

Metrify Health

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Action Plan Summary

The focus of my action plan is regarding the implementation of the AI-enabled precision education platform for academia, specifically for Nurse Practitioner (NP) training. There is a current deficit of physicians in the US, with the Association of American Medical Colleges (AAMC) revealing a need for more than 202,000 physicians to achieve health equity today. Additionally, less medical school students are interested in primary care work, reimbursements have stagnated for physicians, and Family Medicine residencies are closing such as the one at University Hospitals Cleveland Medical Center. Alternatives must be considered to provide health access to underserved and rural communities.

One solution is to strengthen the workforce with NPs. NPs are one of the fastest growing professions according to the Bureau of Labor Statistics (BLS) growing at a 10-year growth rate of 46%. The American Association of Nurse Practitioners (AANP) share that there are over 385,000 active NPs today. What would help is if NPs had Full Practice Authority (FPA) across all 50 states. Currently, NPs must work under a collaborative agreement with physicians. This can limit the location NPs work and can make it difficult for continuity in services if the physician chooses to move or retire. So far, 27 states have passed some level of FPA. But this is not enough.

An AI-enabled precision education platform must be tailored to NP education and clinical work. Having worked to build this for the last 3 years, I am sharing my findings below along with the benefits for health systems, insurance companies, and US citizens.

Why we need an AI-enabled platform for NPs

NP education is condensed into 3 years with a mix of didactic and clinical training that includes simulation labs and clinical rotations with physicians and other NPs. However, NP students, different from medical school students, come in with clinical experience as they were registered nurses (RNs) prior to entering into the NP program. One comment that has been repeated to me is that there is a need to change the mindset from “taking order” to “giving orders”. The change is from relying on a provider’s recommended treatment protocol to making a differential diagnosis and producing a treatment plan as an NP.

An AI-enabled precision education platform tailored for NP education can provide education on developing a differential diagnosis. This is an incredibly important step as it

sets the tone on the next validation steps, including ordering labs, deciding on general palliative or pain-management treatments, considering specialty consults, and developing treatment plans.

Is Copyright data needed to finetune LLM

The benefit of this platform is that most of the work can utilize open-source clinical practice guidelines. Additionally, partnerships with each school can provide access to documents and patient scenarios that are proprietary to each NP program.

The optimal situation will be to access published medical textbook and exercise data. This does not require government intervention to convince publishers to open access to their data. Rather, partnerships with publishing companies can include some form of profit-sharing or equity offer to get access to updated medical education data. This will be an incredible advantage to the care of patients as this tool will provide two main benefits:

- 1) *Standardization of care across all states*
 - a. This platform can consolidate the data across all partnered publishers and optimize the selection for the differential diagnosis and treatment protocols. This can allow for the general understanding of care across all spectrums and provide a tool that can be used in any part of the country and abroad.
- 2) *Geographic-specific care:*
 - a. There are geographic-specific ailments such as Lyme disease and Rocky Mountain Spotted Fever. By utilizing University and University Hospital data, the model can be finetuned to recognize geographic patterns and tailor the education in that manner
 - b. This will further encourage those students to stay in their local region to practice the medicine that they learned for the benefit of the local patient population

Benefits to the Federal Government and Payors

Utilizing NPs will make healthcare more affordable. Studies have shown that preventative care and wellness visits can lead to overall savings. That is why payors push to get physicians to see their patients with a checklist of items on their Annual Wellness Visit (AWV). The current problem is that there are healthcare deserts in addition to a shortage of physicians across the country.

The solution is to utilize NPs. Research has shown that NPs work in rural and underserved communities more than physicians within states that have given NPs Full Practice Authority. Additionally, telemedicine is a great way to reach patients in these communities. Already telemedicine companies are hiring NPs to reach out to patients and have increased access.

An increase in access to providers will lead to frequent wellness visits and a decrease in high cost procedures due to preventable diseases like diabetes.

Benefits to Health Systems

Across the country, health systems and hospitals are failing due to high costs and low reimbursements. By focusing on the utilization of NPs, health systems can save on salary expenses and increase revenue generation. According to the Bureau of Labor Statistics, NPs salary is 60% of Family Physicians and 51% of Physicians in general. These numbers translate to incredible savings.

- 1) *NP salaries are lower and provide cost savings*
 - a. According to the Bureau of Labor Statistics, NPs salary is 60% of Family Physicians and 51% of Physicians in general. These numbers translate to incredible savings to just maintain the existing patient panel size. Even with strong bonus structures for NPs, the overall cost would still be lower than a physician-only organization.
- 2) *Hiring NPs can generate more revenue*
 - a. If the same salary expenditures were used to hire NPs, a health system could hire more NPs than they could Primary Care Physicians. If each NP could maintain the same patient panel size as their physician counterpart, this transition would translate to an overall greater patient volume leading to greater revenue generation. More patients lead to more revenue even in a capitated model.
- 3) *Bringing in NPs can increase patient satisfaction*
 - a. Patients are currently struggling with wait times with studies showing an average wait time of 26 days to see a physician
 - b. More NPs can decrease the wait time considerably allowing patients to see a provider and even get the necessary referral to see a specialist. Decreasing wait times will translate to great patient satisfaction as they can access their provider more frequently.
 - c. If this helps with a higher star rating, the benefits spill over to greater revenue generation as well.

Benefits to US citizens

Even with our technology today, there are millions of citizens who cannot get access to a physician. Moreso, family practices are closing down in rural America as reimbursement rates stagnate for physicians. Research has identified that NPs work in rural and underserved communities more than physicians within states that have given NPs Full Practice Authority. Additionally, telemedicine is a great way to reach patients in these communities. Already telemedicine companies are hiring NPs to reach out to patients and have increased access.

An increase in access to providers will allow for frequent wellness visits and a decrease in high cost procedures due to preventable diseases like diabetes. This will translate to cost savings for the patient who lacks coverage or does not have a strong insurance plan.

Additionally, NPs trained in mental health care can increase access to behavioral health treatments that are increasingly needed throughout the country.

Call to Action

- 1) NP Full Practice Authority must be provided in all 50 states
- 2) AI-enabled Precision Education and Precision Healthcare must be provided to NPs
- 3) This opportunity can be leveraged by other professionals like Physicians Assistants, Nurses, and Lawyers.
- 4) This will lead to more primary care providers and trickle down to allow physicians to focus their education on secondary, tertiary, and quaternary care.